HDOC DAY-50

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BATCH-15

Question

Write a C program to read the size and elements of a variable-length integer array nums[]. Pass the array to a function majority_element() using pointers. Find and display the element that occurs more than size/2 times. Display -1 if no majority element exists.

Aim

To write a C program to find the majority element in an array using functions and pointers.

Algorithm

Step 1: Start.

Step 2: Read the size n of the array.

Step 3: Read n array elements.

Step 4: Pass the array address to majority_element().

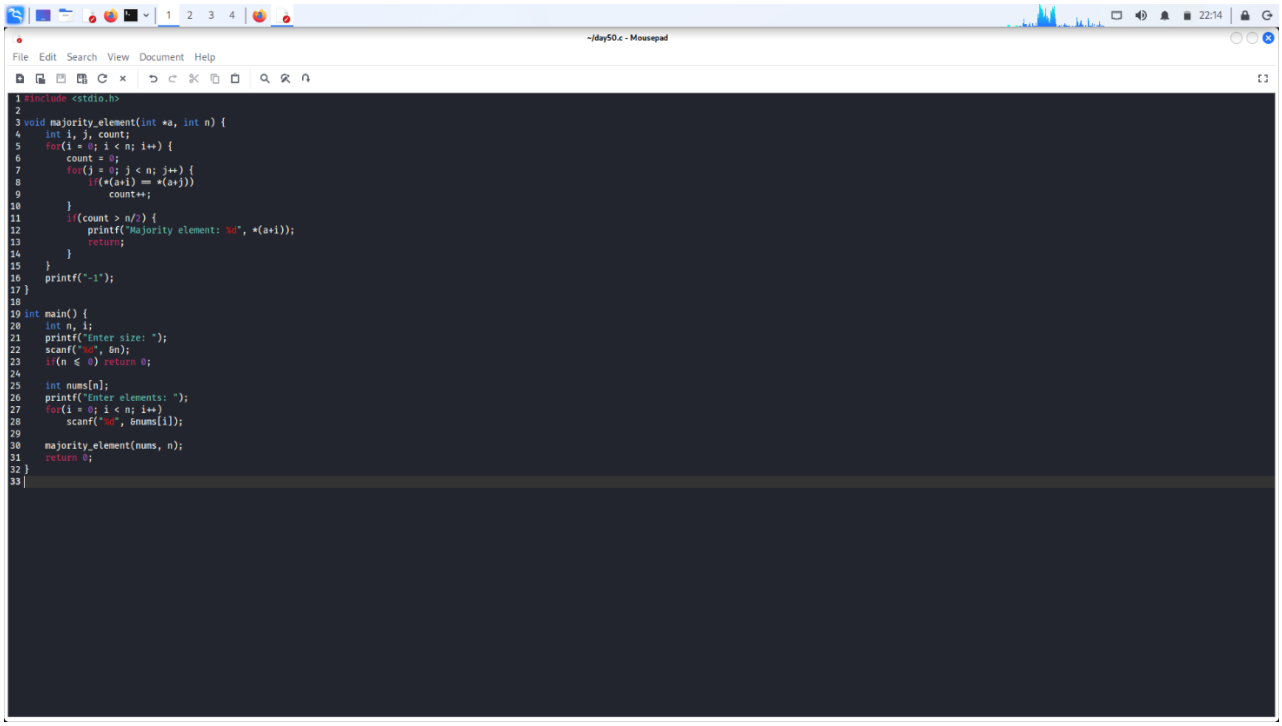
Step 5: Count the occurrences of each element using nested loops.

Step 6: If the count is greater than n/2, print the majority element.

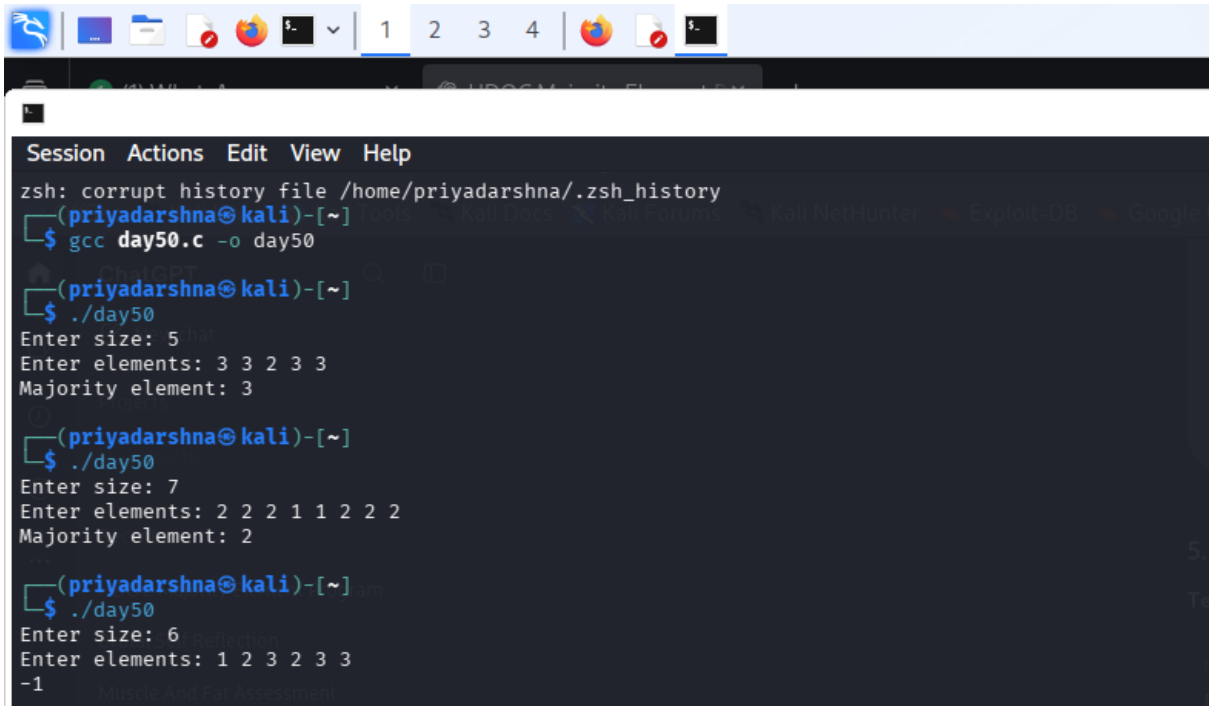
Step 7: Otherwise, print -1.

Step 8: Stop.

>C PROGRAM:



>OUTPUT:


1 2 3 4
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna㉿kali)-[~]
$ gcc day50.c -o day50
(priyadarshna㉿kali)-[~]
$ ./day50
Enter size: 5
Enter elements: 3 3 2 3 3
Majority element: 3
(priyadarshna㉿kali)-[~]
$ ./day50
Enter size: 7
Enter elements: 2 2 2 1 1 2 2 2
Majority element: 2
(priyadarshna㉿kali)-[~]
$ ./day50
Enter size: 6
Enter elements: 1 2 3 2 3 3
-1